

PM 566 Homework 1

AUTHOR

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Library Packages

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
library(maps)
library(leaflet)
library(data.table)
```

Attaching package: 'data.table'

The following objects are masked from 'package:dplyr':

between, first, last

```
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ forcats	1.0.0	✓ stringr	1.5.2
✓ lubridate	1.9.4	✓ tibble	3.3.0
✓ purrr	1.1.0	✓ tidyr	1.3.1
✓ readr	2.1.5		

— Conflicts — tidyverse_conflicts() —

* data.table::between()	masks	dplyr::between()
* dplyr::filter()	masks	stats::filter()
* data.table::first()	masks	dplyr::first()
* lubridate::hour()	masks	data.table::hour()
* lubridate::isoweek()	masks	data.table::isoweek()
* dplyr::lag()	masks	stats::lag()

```

* data.table::last()      masks dplyr::last()
* purrr::map()            masks maps::map()
* lubridate::mday()       masks data.table::mday()
* lubridate::minute()     masks data.table::minute()
* lubridate::month()      masks data.table::month()
* lubridate::quarter()    masks data.table::quarter()
* lubridate::second()     masks data.table::second()
* purrr::transpose()      masks data.table::transpose()
* lubridate::wday()        masks data.table::wday()
* lubridate::week()        masks data.table::week()
* lubridate::yday()        masks data.table::yday()
* lubridate::year()        masks data.table::year()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```

Question 1. (30 points) Given the formulated question from the assignment description, you will now conduct EDA Checklist items 1-5. First, download 2002 and 2022 data for all sites in California from the EPA Air Quality Data website, then read the data into R. For each of the two datasets, check the dimensions, headers, footers, variable names and variable types. Check the distribution of the key variable we are analyzing (PM). Write up a summary of all of your findings.

```

# Import data
eda02 = read.csv("/Users/ldh/Library/Mobile Documents/com~apple~CloudDocs/USC PhD/STUDY D
eda22 = read.csv("/Users/ldh/Library/Mobile Documents/com~apple~CloudDocs/USC PhD/STUDY D

```

```

# check the dimensions, headers, footers, variable names and variable type for 2002 data
dim(eda02) # dimensions

```

```
[1] 15976    22
```

```
head(eda02) # headers
```

	Date	Source	Site.ID	POC	Daily.Mean.PM2.5.Concentration	Units
1	01/05/2002	AQS	60010007	1	25.1	ug/m3 LC
2	01/06/2002	AQS	60010007	1	31.6	ug/m3 LC
3	01/08/2002	AQS	60010007	1	21.4	ug/m3 LC
4	01/11/2002	AQS	60010007	1	25.9	ug/m3 LC
5	01/14/2002	AQS	60010007	1	34.5	ug/m3 LC
6	01/17/2002	AQS	60010007	1	41.0	ug/m3 LC

	Daily.AQI.Value	Local.Site.Name	Daily.Obs.Count	Percent.Complete
1	81	Livermore	1	100
2	93	Livermore	1	100
3	74	Livermore	1	100
4	82	Livermore	1	100
5	98	Livermore	1	100
6	115	Livermore	1	100

	AQS.Parameter.Code	AQS.Parameter.Description	Method.Code
1	88101	PM2.5 – Local Conditions	120
2	88101	PM2.5 – Local Conditions	120
3	88101	PM2.5 – Local Conditions	120
4	88101	PM2.5 – Local Conditions	120
5	88101	PM2.5 – Local Conditions	120
6	88101	PM2.5 – Local Conditions	120

	Method.Description	CBSA.Code
1	Andersen RAAS2.5–300 PM2.5 SEQ w/WINS	41860
2	Andersen RAAS2.5–300 PM2.5 SEQ w/WINS	41860
3	Andersen RAAS2.5–300 PM2.5 SEQ w/WINS	41860
4	Andersen RAAS2.5–300 PM2.5 SEQ w/WINS	41860
5	Andersen RAAS2.5–300 PM2.5 SEQ w/WINS	41860
6	Andersen RAAS2.5–300 PM2.5 SEQ w/WINS	41860

	CBSA.Name	State.FIPS.Code	State	County.FIPS.Code
1	San Francisco–Oakland–Hayward, CA	6	California	1
2	San Francisco–Oakland–Hayward, CA	6	California	1
3	San Francisco–Oakland–Hayward, CA	6	California	1
4	San Francisco–Oakland–Hayward, CA	6	California	1
5	San Francisco–Oakland–Hayward, CA	6	California	1
6	San Francisco–Oakland–Hayward, CA	6	California	1

	County	Site.Latitude	Site.Longitude
1	Alameda	37.68753	–121.7842
2	Alameda	37.68753	–121.7842
3	Alameda	37.68753	–121.7842
4	Alameda	37.68753	–121.7842
5	Alameda	37.68753	–121.7842
6	Alameda	37.68753	–121.7842

```
tail(eda02) # footers
```

	Date	Source	Site.ID	POC	Daily.Mean.PM2.5.Concentration	Units
15971	12/10/2002	AQS	61131003	1	15	ug/m3 LC
15972	12/13/2002	AQS	61131003	1	15	ug/m3 LC
15973	12/22/2002	AQS	61131003	1	1	ug/m3 LC
15974	12/25/2002	AQS	61131003	1	23	ug/m3 LC
15975	12/28/2002	AQS	61131003	1	5	ug/m3 LC
15976	12/31/2002	AQS	61131003	1	6	ug/m3 LC

	Daily.AQI.Value	Local.Site.Name	Daily.Obs.Count	Percent.Complete
15971	62	Woodland–Gibson Road	1	100
15972	62	Woodland–Gibson Road	1	100
15973	6	Woodland–Gibson Road	1	100
15974	77	Woodland–Gibson Road	1	100

15975	28 Woodland-Gibson Road	1	100
15976	33 Woodland-Gibson Road	1	100
	AQS.Parameter.Code	AQS.Parameter.Description	Method.Code
15971	88101	PM2.5 - Local Conditions	117
15972	88101	PM2.5 - Local Conditions	117
15973	88101	PM2.5 - Local Conditions	117
15974	88101	PM2.5 - Local Conditions	117
15975	88101	PM2.5 - Local Conditions	117
15976	88101	PM2.5 - Local Conditions	117
	Method.Description	CBSA.Code	
15971	R & P Model 2000 PM2.5 Sampler w/WINS	40900	
15972	R & P Model 2000 PM2.5 Sampler w/WINS	40900	
15973	R & P Model 2000 PM2.5 Sampler w/WINS	40900	
15974	R & P Model 2000 PM2.5 Sampler w/WINS	40900	
15975	R & P Model 2000 PM2.5 Sampler w/WINS	40900	
15976	R & P Model 2000 PM2.5 Sampler w/WINS	40900	
	CBSA.Name	State.FIPS.Code	State
15971	Sacramento--Roseville--Arden-Arcade, CA	6	California
15972	Sacramento--Roseville--Arden-Arcade, CA	6	California
15973	Sacramento--Roseville--Arden-Arcade, CA	6	California
15974	Sacramento--Roseville--Arden-Arcade, CA	6	California
15975	Sacramento--Roseville--Arden-Arcade, CA	6	California
15976	Sacramento--Roseville--Arden-Arcade, CA	6	California
	County.FIPS.Code	County	Site.Latitude Site.Longitude
15971	113	Yolo	38.66121 -121.7327
15972	113	Yolo	38.66121 -121.7327
15973	113	Yolo	38.66121 -121.7327
15974	113	Yolo	38.66121 -121.7327
15975	113	Yolo	38.66121 -121.7327
15976	113	Yolo	38.66121 -121.7327

```
names(eda02) # variables name
```

```
[1] "Date"
[3] "Site.ID"
[5] "Daily.Mean.PM2.5.Concentration"
[7] "Daily.AQI.Value"
[9] "Daily.Obs.Count"
[11] "AQS.Parameter.Code"
[13] "Method.Code"
[15] "CBSA.Code"
[17] "State.FIPS.Code"
[19] "County.FIPS.Code"
[21] "Site.Latitude"

"Source"
"POC"
"Units"
"Local.Site.Name"
"Percent.Complete"
"AQS.Parameter.Description"
"Method.Description"
"CBSA.Name"
"State"
"County"
"Site.Longitude"
```

```
str(eda02) # type
```

```
'data.frame': 15976 obs. of 22 variables:
 $ Date : chr "01/05/2002" "01/06/2002" "01/08/2002"
```

```

"01/11/2002" ...
$ Source           : chr  "AQS" "AQS" "AQS" "AQS" ...
$ Site.ID          : int   60010007 60010007 60010007 60010007 60010007
60010007 60010007 60010007 60010007 60010007 ...
$ POC              : int   1 1 1 1 1 1 1 1 1 1 ...
$ Daily.Mean.PM2.5.Concentration: num  25.1 31.6 21.4 25.9 34.5 41 29.3 15 18.8 37.9 ...
$ Units            : chr   "ug/m3 LC" "ug/m3 LC" "ug/m3 LC" "ug/m3 LC" ...
$ Daily.AQI.Value   : int   81 93 74 82 98 115 89 62 69 107 ...
$ Local.Site.Name   : chr   "Livermore" "Livermore" "Livermore" "Livermore"
...
$ Daily.Obs.Count   : int   1 1 1 1 1 1 1 1 1 1 ...
$ Percent.Complete  : num   100 100 100 100 100 100 100 100 100 100 ...
$ AQS.Parameter.Code : int   88101 88101 88101 88101 88101 88101 88101 88101
88101 88101 ...
$ AQS.Parameter.Description : chr  "PM2.5 - Local Conditions" "PM2.5 - Local
Conditions" "PM2.5 - Local Conditions" "PM2.5 - Local Conditions" ...
$ Method.Code       : int   120 120 120 120 120 120 120 120 120 120 ...
$ Method.Description : chr   "Andersen RAAS2.5-300 PM2.5 SEQ w/WINS" "Andersen
RAAS2.5-300 PM2.5 SEQ w/WINS" "Andersen RAAS2.5-300 PM2.5 SEQ w/WINS" "Andersen RAAS2.5-
300 PM2.5 SEQ w/WINS" ...
$ CBSA.Code         : int   41860 41860 41860 41860 41860 41860 41860 41860
41860 41860 ...
$ CBSA.Name         : chr   "San Francisco-Oakland-Hayward, CA" "San
San Francisco-Oakland-Hayward, CA" "San Francisco-Oakland-Hayward, CA" "San Francisco-
Oakland-Hayward, CA" ...
$ State.FIPS.Code   : int   6 6 6 6 6 6 6 6 6 6 ...
$ State             : chr   "California" "California" "California"
"California" ...
$ County.FIPS.Code  : int   1 1 1 1 1 1 1 1 1 1 ...
$ County            : chr   "Alameda" "Alameda" "Alameda" "Alameda" ...
$ Site.Latitude     : num   37.7 37.7 37.7 37.7 37.7 ...
$ Site.Longitude    : num   -122 -122 -122 -122 -122 ...

```

```

# check the dimensions, headers, footers, variable names and variable type for 2022 data
dim(eda22)

```

```
[1] 59918    22
```

```
head(eda22)
```

	Date	Source	Site.ID	POC	Daily.Mean.PM2.5.Concentration	Units
1	01/01/2022	AQS	60010007	3	12.7	ug/m3 LC
2	01/02/2022	AQS	60010007	3	13.9	ug/m3 LC
3	01/03/2022	AQS	60010007	3	7.1	ug/m3 LC
4	01/04/2022	AQS	60010007	3	3.7	ug/m3 LC
5	01/05/2022	AQS	60010007	3	4.2	ug/m3 LC
6	01/06/2022	AQS	60010007	3	3.8	ug/m3 LC
	Daily.AQI.Value		Local.Site.Name		Daily.Obs.Count	
1	58		Livermore		100	

2	60	Livermore	1	100
3	39	Livermore	1	100
4	21	Livermore	1	100
5	23	Livermore	1	100
6	21	Livermore	1	100

AQS.Parameter.Code	AQS.Parameter.Description	Method.Code
1	88101 PM2.5 – Local Conditions	170
2	88101 PM2.5 – Local Conditions	170
3	88101 PM2.5 – Local Conditions	170
4	88101 PM2.5 – Local Conditions	170
5	88101 PM2.5 – Local Conditions	170
6	88101 PM2.5 – Local Conditions	170

Method.Description	CBSA.Code
1 Met One BAM-1020 Mass Monitor w/VSCC	41860
2 Met One BAM-1020 Mass Monitor w/VSCC	41860
3 Met One BAM-1020 Mass Monitor w/VSCC	41860
4 Met One BAM-1020 Mass Monitor w/VSCC	41860
5 Met One BAM-1020 Mass Monitor w/VSCC	41860
6 Met One BAM-1020 Mass Monitor w/VSCC	41860

CBSA.Name	State.FIPS.Code	State	County.FIPS.Code
1 San Francisco–Oakland–Hayward, CA	6	California	1
2 San Francisco–Oakland–Hayward, CA	6	California	1
3 San Francisco–Oakland–Hayward, CA	6	California	1
4 San Francisco–Oakland–Hayward, CA	6	California	1
5 San Francisco–Oakland–Hayward, CA	6	California	1
6 San Francisco–Oakland–Hayward, CA	6	California	1

County	Site.Latitude	Site.Longitude
1 Alameda	37.68753	–121.7842
2 Alameda	37.68753	–121.7842
3 Alameda	37.68753	–121.7842
4 Alameda	37.68753	–121.7842
5 Alameda	37.68753	–121.7842
6 Alameda	37.68753	–121.7842

```
tail(eda22)
```

Date	Source	Site.ID	POC	Daily.Mean.PM2.5.Concentration	Units
59913	12/01/2022	AQS 61131003	1	3.4	ug/m3 LC
59914	12/07/2022	AQS 61131003	1	3.8	ug/m3 LC
59915	12/13/2022	AQS 61131003	1	6.0	ug/m3 LC
59916	12/19/2022	AQS 61131003	1	34.8	ug/m3 LC
59917	12/25/2022	AQS 61131003	1	23.2	ug/m3 LC
59918	12/31/2022	AQS 61131003	1	1.0	ug/m3 LC

Daily.AQI.Value	Local.Site.Name	Daily.Obs.Count	Percent.Complete
59913	19 Woodland–Gibson Road	1	100
59914	21 Woodland–Gibson Road	1	100
59915	33 Woodland–Gibson Road	1	100
59916	99 Woodland–Gibson Road	1	100
59917	77 Woodland–Gibson Road	1	100
59918	6 Woodland–Gibson Road	1	100

	AQS.Parameter.Code	AQS.Parameter.Description	Method.Code	
59913	88101	PM2.5 – Local Conditions	145	
59914	88101	PM2.5 – Local Conditions	145	
59915	88101	PM2.5 – Local Conditions	145	
59916	88101	PM2.5 – Local Conditions	145	
59917	88101	PM2.5 – Local Conditions	145	
59918	88101	PM2.5 – Local Conditions	145	
		Method.Description	CBSA.Code	
59913	R & P Model 2025	PM-2.5 Sequential Air Sampler w/VSCC	40900	
59914	R & P Model 2025	PM-2.5 Sequential Air Sampler w/VSCC	40900	
59915	R & P Model 2025	PM-2.5 Sequential Air Sampler w/VSCC	40900	
59916	R & P Model 2025	PM-2.5 Sequential Air Sampler w/VSCC	40900	
59917	R & P Model 2025	PM-2.5 Sequential Air Sampler w/VSCC	40900	
59918	R & P Model 2025	PM-2.5 Sequential Air Sampler w/VSCC	40900	
	CBSA.Name	State.FIPS.Code	State	
59913	Sacramento--Roseville--Arden-Arcade, CA	6	California	
59914	Sacramento--Roseville--Arden-Arcade, CA	6	California	
59915	Sacramento--Roseville--Arden-Arcade, CA	6	California	
59916	Sacramento--Roseville--Arden-Arcade, CA	6	California	
59917	Sacramento--Roseville--Arden-Arcade, CA	6	California	
59918	Sacramento--Roseville--Arden-Arcade, CA	6	California	
	County.FIPS.Code	County	Site.Latitude	Site.Longitude
59913	113	Yolo	38.66121	-121.7327
59914	113	Yolo	38.66121	-121.7327
59915	113	Yolo	38.66121	-121.7327
59916	113	Yolo	38.66121	-121.7327
59917	113	Yolo	38.66121	-121.7327
59918	113	Yolo	38.66121	-121.7327

```
names(eda22)
```

```
[1] "Date"                "Source"
[3] "Site.ID"             "POC"
[5] "Daily.Mean.PM2.5.Concentration" "Units"
[7] "Daily.AQI.Value"     "Local.Site.Name"
[9] "Daily.Obs.Count"     "Percent.Complete"
[11] "AQS.Parameter.Code"  "AQS.Parameter.Description"
[13] "Method.Code"         "Method.Description"
[15] "CBSA.Code"           "CBSA.Name"
[17] "State.FIPS.Code"     "State"
[19] "County.FIPS.Code"    "County"
[21] "Site.Latitude"       "Site.Longitude"
```

```
str(eda22)
```

```
'data.frame': 59918 obs. of 22 variables:
 $ Date                : chr "01/01/2022" "01/02/2022" "01/03/2022"
"01/04/2022" ...
 $ Source              : chr "AQS" "AQS" "AQS" "AQS" ...
```

```

$ Site.ID : int 60010007 60010007 60010007 60010007 60010007
60010007 60010007 60010007 60010007 60010007 ...
$ POC : int 3 3 3 3 3 3 3 3 3 3 ...
$ Daily.Mean.PM2.5.Concentration: num 12.7 13.9 7.1 3.7 4.2 3.8 2.3 6.9 13.6 11.2 ...
$ Units : chr "ug/m3 LC" "ug/m3 LC" "ug/m3 LC" "ug/m3 LC" ...
$ Daily.AQI.Value : int 58 60 39 21 23 21 13 38 59 55 ...
$ Local.Site.Name : chr "Livermore" "Livermore" "Livermore" "Livermore"
...
$ Daily.Obs.Count : int 1 1 1 1 1 1 1 1 1 1 ...
$ Percent.Complete : num 100 100 100 100 100 100 100 100 100 100 ...
$ AQS.Parameter.Code : int 88101 88101 88101 88101 88101 88101 88101 88101
88101 88101 ...
$ AQS.Parameter.Description : chr "PM2.5 - Local Conditions" "PM2.5 - Local
Conditions" "PM2.5 - Local Conditions" "PM2.5 - Local Conditions" ...
$ Method.Code : int 170 170 170 170 170 170 170 170 170 170 ...
$ Method.Description : chr "Met One BAM-1020 Mass Monitor w/VSCC" "Met One
BAM-1020 Mass Monitor w/VSCC" "Met One BAM-1020 Mass Monitor w/VSCC" "Met One BAM-1020
Mass Monitor w/VSCC" ...
$ CBSA.Code : int 41860 41860 41860 41860 41860 41860 41860 41860
41860 41860 ...
$ CBSA.Name : chr "San Francisco-Oakland-Hayward, CA" "San
Francisco-Oakland-Hayward, CA" "San Francisco-Oakland-Hayward, CA" "San Francisco-
Oakland-Hayward, CA" ...
$ State.FIPS.Code : int 6 6 6 6 6 6 6 6 6 6 ...
$ State : chr "California" "California" "California"
"California" ...
$ County.FIPS.Code : int 1 1 1 1 1 1 1 1 1 1 ...
$ County : chr "Alameda" "Alameda" "Alameda" "Alameda" ...
$ Site.Latitude : num 37.7 37.7 37.7 37.7 37.7 ...
$ Site.Longitude : num -122 -122 -122 -122 -122 ...

```

```

# Check the distribution of the key variable we are analyzing (PM 2.5)
summary(eda02$Daily.Mean.PM2.5.Concentration)

```

```

Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00 7.00 12.00 16.12 20.50 104.30

```

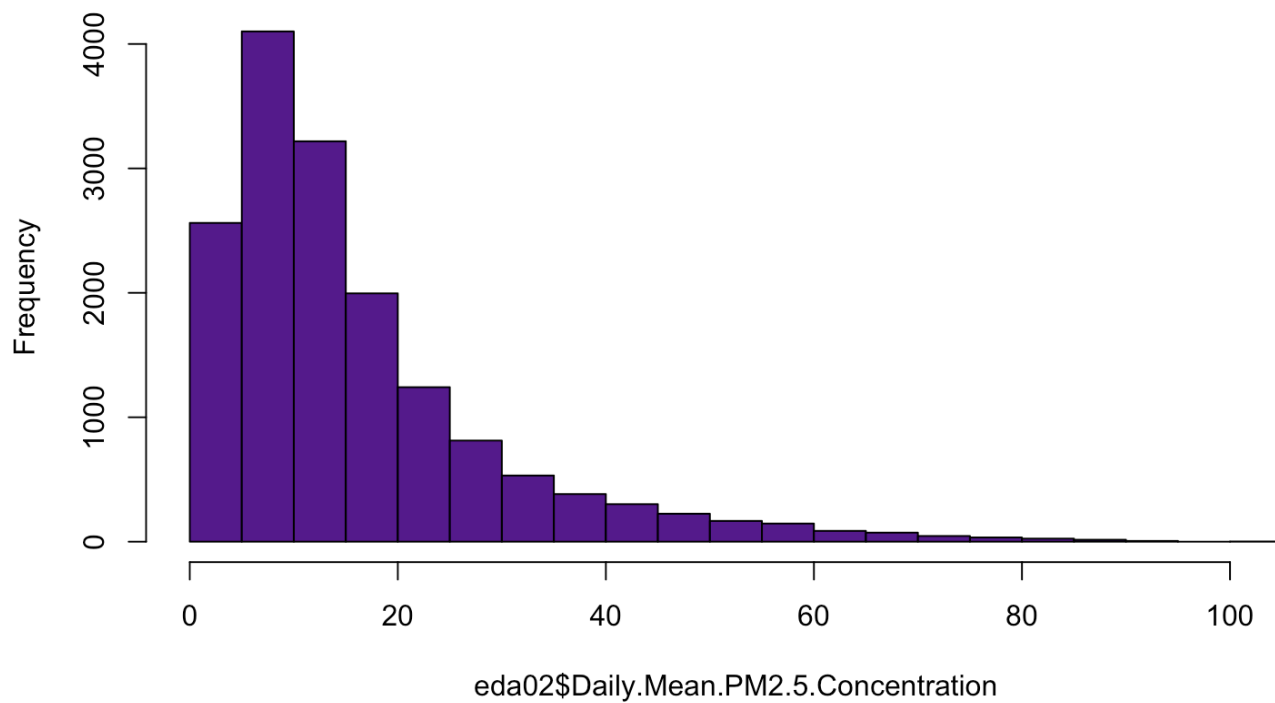
```
summary(eda22$Daily.Mean.PM2.5.Concentration)
```

```

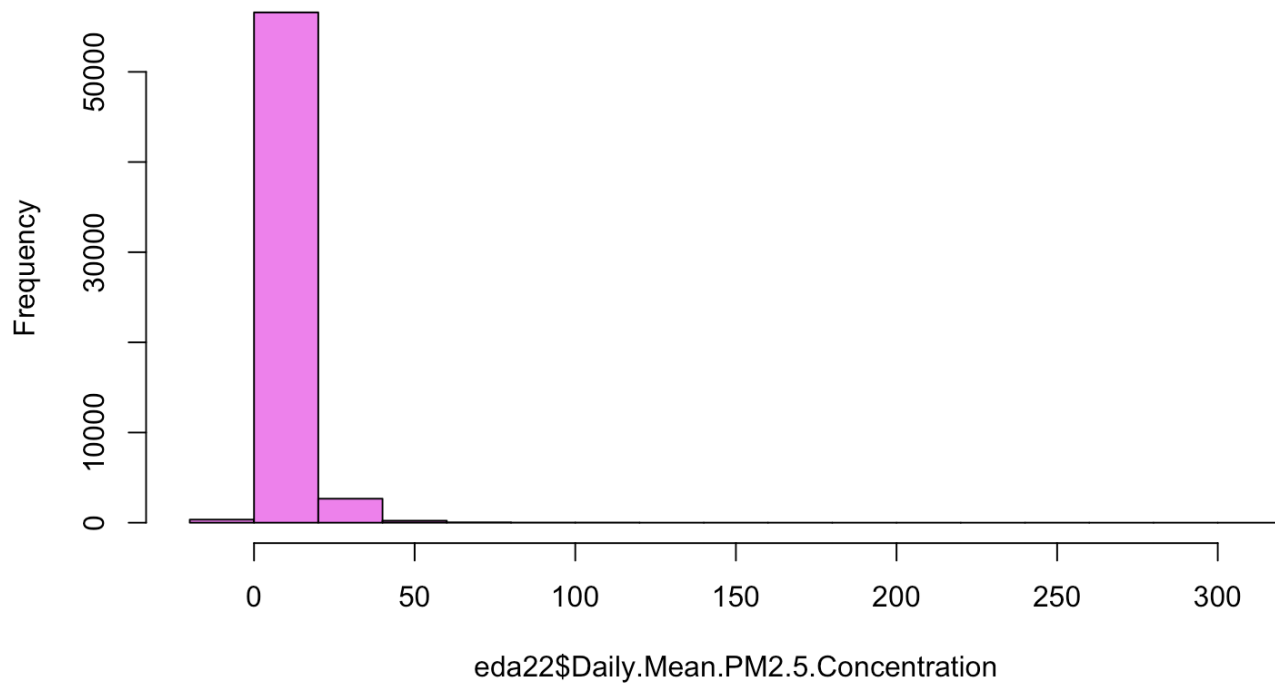
Min. 1st Qu. Median Mean 3rd Qu. Max.
-6.700 4.100 6.800 8.414 10.700 302.500

```

```
hist(eda02$Daily.Mean.PM2.5.Concentration, col = "purple4")
```


Histogram of eda02\$Daily.Mean.PM2.5.Concentration

```
hist(eda22$Daily.Mean.PM2.5.Concentration, col = "violet")
```

Histogram of eda22\$Daily.Mean.PM2.5.Concentration

Summary:The 2002 dataset had 15,976 observations and 22 variables, whereas the 2022 dataset included 59,918 observations and 22 variables. The distribution of the main variable PM 2.5 in the 2002 dataset is as follows: Minimum = 0.00, 1st Quartile = 7.00, Median = 12.00, Mean = 16.12, 3rd Quartile = 20.50, Maximum = 104.30. The histogram exhibited a right-skewed distribution, with the majority of daily PM2.5 values ranging from 5 to 25 $\mu\text{g}/\text{m}^3$, accompanied by a lengthy tail extending to around 100 $\mu\text{g}/\text{m}^3$. Regarding the 2022 dataset, the Minimum is -6.70, the 1st Quartile is 4.10, the Median is 6.80, the Mean is 8.41, the 3rd Quartile is 10.70, and the Maximum is 302.50. The histogram indicated that most results were concentrated below 20 $\mu\text{g}/\text{m}^3$, with very high values signifying few occurrences. The distribution is right-skewed; nonetheless, the total concentrations are much lower than in 2002. The overall trend of PM2.5 concentrations significantly decreased from 2002 to 2022. The average PM2.5 levels decreased from 16.1 $\mu\text{g}/\text{m}^3$ in 2002 to 8.4 $\mu\text{g}/\text{m}^3$ in 2022, while the median values fell from 12.0 $\mu\text{g}/\text{m}^3$ to 6.8 $\mu\text{g}/\text{m}^3$. The 2022 dataset includes more severe maximum values (302.5 $\mu\text{g}/\text{m}^3$) and an extreme minimum value ($<0 \mu\text{g}/\text{m}^3$)

Question 2 (10 points) Combine the two years of data into one data frame. Use the Date variable to create a new column for year, which will serve as an identifier. Change the names of the key variables so that they are easier to refer to in your code.

```
# check on NA for the variable Date  
sum(is.na(eda02$Date))
```

```
[1] 0
```

```
sum(is.na(eda22$Date))
```

```
[1] 0
```

```
# Combine the two data into one
eda_all <- merge(x=eda02, y=eda22, all=TRUE)
```

```
# check NA of date variable in the new combined dataset
sum(is.na(eda_all$Date))
```

```
[1] 0
```

```
# convert variable Date into date form variable
eda_all$Date <- as.Date(eda_all$Date, format = "%m/%d/%Y")
```

```
# Use the Date variable to create a new column for year (based on the year information of
eda_all$year <- format(eda_all$Date, "%Y")
table(eda_all$year) # check on the accuracy of the variable year
```

```
2002  2022
15976 59918
```

```
# rename the key variable "Daily.Mean.PM2.5.Concentration"
names(eda_all)[names(eda_all) == "Daily.Mean.PM2.5.Concentration"] <- "pm2.5_avg_per_day"
```

Question 3 (20 points) Create a basic map (or maps) in leaflet showing the locations of the monitoring sites, using different colors for each year. Summarize the spatial distribution of the sites. Does this distribution change from 2002 to 2022?

```
# ggplot approach
# 1. Keep one row per site per year (otherwise you'll plot every daily record)
sites <- eda_all %>%
  select(Site.ID, year, Site.Latitude, Site.Longitude) %>%
  filter(!is.na(Site.Latitude), !is.na(Site.Longitude)) %>%
  distinct(Site.ID, year, .keep_all = TRUE)

# 2. Basic USA basemap (lower 48)
usa <- map_data("state")

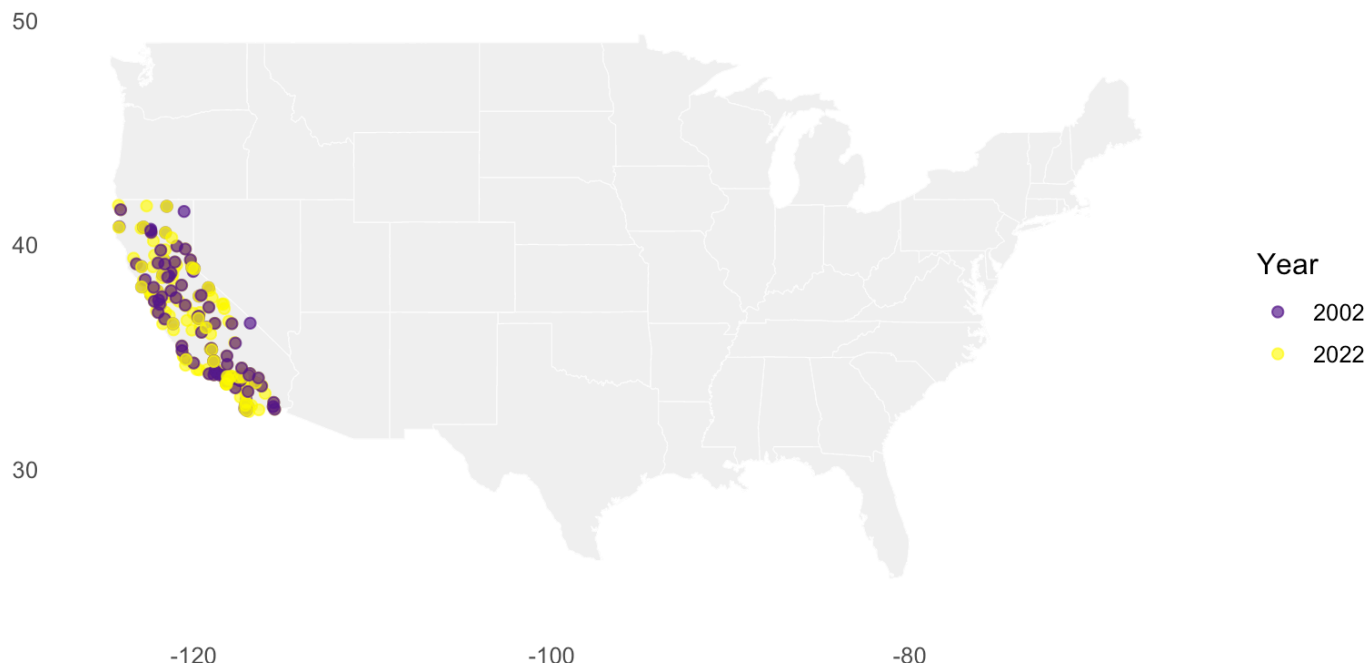
# 3a. One map, points colored by year
ggplot() +
  geom_polygon(data = usa,
    aes(long, lat, group = group),
    fill = "grey95", color = "white", linewidth = 0.2) +
  geom_point(data = sites,
```

```

aes(x = Site.Longitude, y = Site.Latitude, color = factor(year)),
alpha = 0.7, size = 1.8) +
scale_color_manual(values = c("2002" = "purple4", "2022" = "yellow1")) +
coord_quickmap(xlim = c(-125, -66), ylim = c(24, 50)) +
labs(title = "EPA PM2.5 Monitoring Sites (2002 vs 2022)",
x = NULL, y = NULL, color = "Year") +
theme_minimal(base_size = 12) +
theme(panel.grid = element_blank())

```

EPA PM2.5 Monitoring Sites (2002 vs 2022)

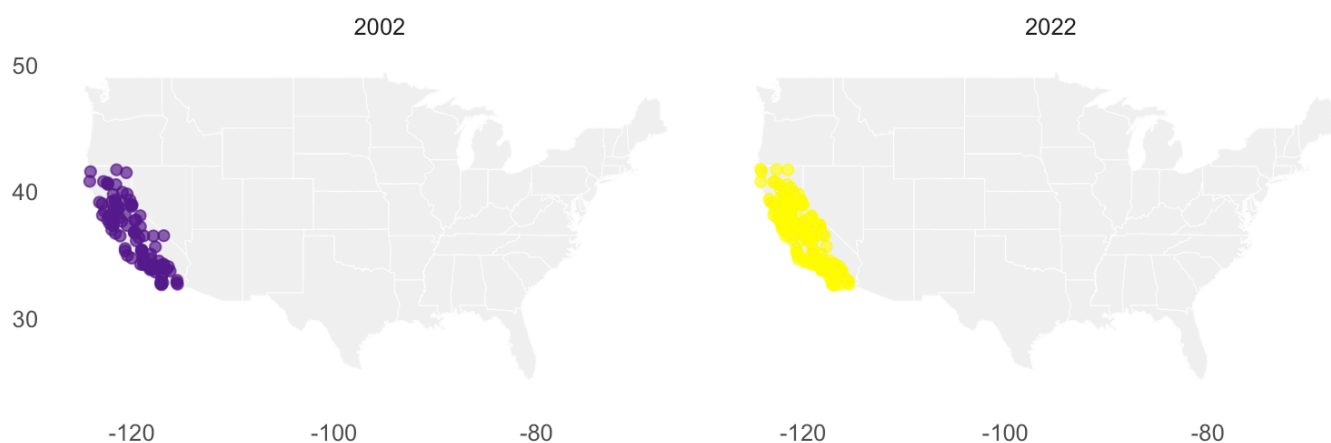


```

# 3b. Faceted version (separate panel for each year)
ggplot() +
  geom_polygon(data = usa,
    aes(long, lat, group = group),
    fill = "grey95", color = "white", linewidth = 0.2) +
  geom_point(data = sites,
    aes(x = Site.Longitude, y = Site.Latitude, color = factor(year)),
    alpha = 0.7, size = 1.8, show.legend = FALSE) +
  scale_color_manual(values = c("2002" = "purple4", "2022" = "yellow1")) +
  coord_quickmap(xlim = c(-125, -66), ylim = c(24, 50)) +
  facet_wrap(~ year) +
  labs(title = "EPA PM2.5 Monitoring Sites by Year",
    x = NULL, y = NULL) +
  theme_minimal(base_size = 12) +
  theme(panel.grid = element_blank())

```

EPA PM2.5 Monitoring Sites by Year



```
# inleaflet approach (for double check the accuracy)

# 1) One row per site x year; keep needed columns; basic QC
sites <- eda_all %>%
  select(Site.ID, year, State, County, Site.Latitude, Site.Longitude) %>%
  filter(!is.na(Site.Latitude), !is.na(Site.Longitude)) %>%
  filter(Site.Latitude >= -90, Site.Latitude <= 90,
         Site.Longitude >= -180, Site.Longitude <= 180) %>%
  distinct(Site.ID, year, .keep_all = TRUE) %>%
  mutate(
    year = factor(year, levels = sort(unique(year))),
    popup = paste0(
      "<b>Site.ID:</b> ", Site.ID,
      "<br><b>Year:</b> ", year,
      "<br><b>State:</b> ", State,
      "<br><b>County:</b> ", County
    )
  )

# (Optional) If many longitudes are positive but should be west longitudes, uncomment:
# sites <- sites %>%
#   mutate(Site.Longitude = ifelse(Site.Longitude > 0 & Site.Longitude <= 180,
#                                   -Site.Longitude, Site.Longitude))
#

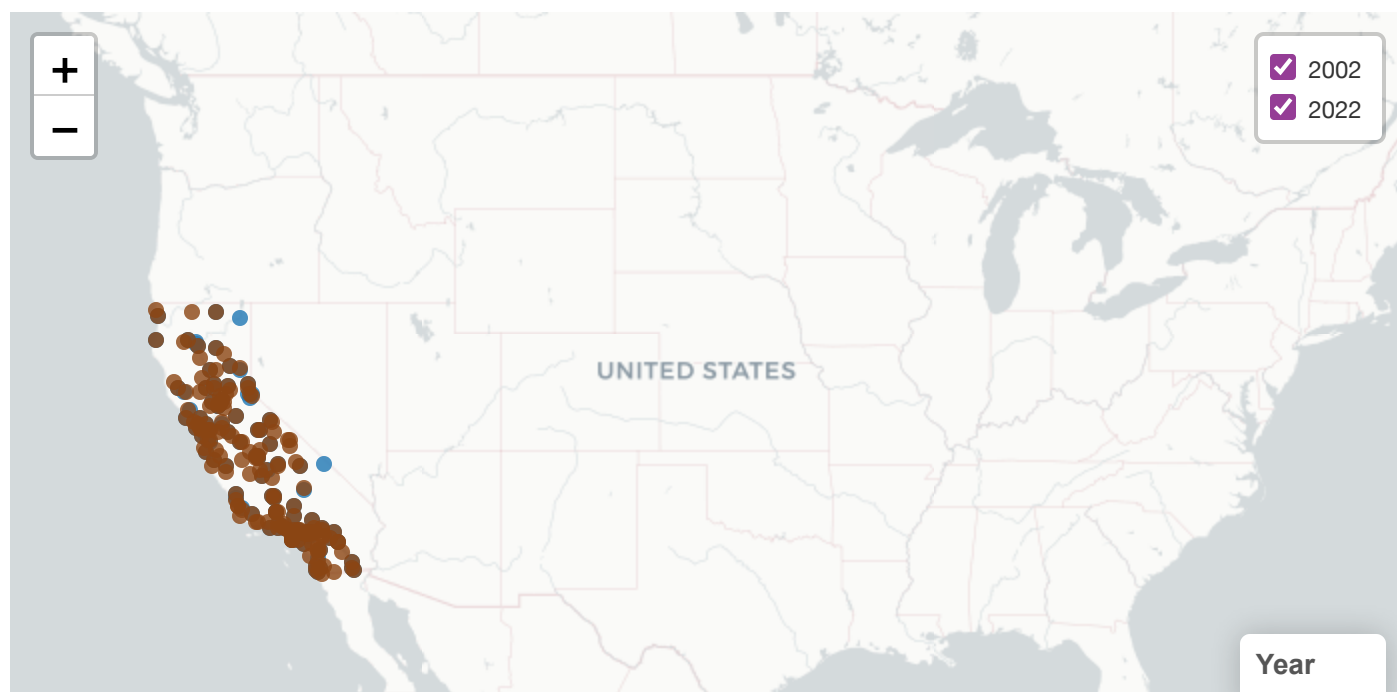
# Color palette by year
pal <- colorFactor(palette = c("#1f77b4", "chocolate4"), domain = sites$year)
```

```
# 2) Build leaflet map with separate layers for each year + layer control
leaflet(options = leafletOptions(minZoom = 2)) %>%
  addProviderTiles(providers$CartoDB.Positron) %>%
  fitBounds(lng1 = -125, lat1 = 24, lng2 = -66, lat2 = 50) %>% # lower 48; adjust or rem

# Layer: 2002
addCircleMarkers(
  data = subset(sites, year == levels(sites$year)[1]),
  lng = ~Site.Longitude, lat = ~Site.Latitude,
  radius = 4, stroke = FALSE, fillOpacity = 0.8,
  fillColor = ~pal(year),
  popup = ~popup,
  group = as.character(levels(sites$year)[1])
) %>%

# Layer: 2022
addCircleMarkers(
  data = subset(sites, year == levels(sites$year)[2]),
  lng = ~Site.Longitude, lat = ~Site.Latitude,
  radius = 4, stroke = FALSE, fillOpacity = 0.8,
  fillColor = ~pal(year),
  popup = ~popup,
  group = as.character(levels(sites$year)[2])
) %>%

# Legend + layer control
addLegend("bottomright", pal = pal, values = sites$year, title = "Year", opacity = 1) %
addLayersControl(
  overlayGroups = as.character(levels(sites$year)),
  options = layersControlOptions(collapsed = FALSE)
)
```





Summarization: The spatial distribution of EPA PM2.5 monitoring sites in 2002 and 2022 is mostly located in California, particularly along the west coast, with an extension inland into the middle and southern areas of the state. The sites are centered in major urban regions. A concentrated cluster of monitors is present in coastal and metropolitan regions, indicating a focus on areas with high population density and air pollution concerns.

Change of distribution: A substantial change in distribution occurred between 2002 and 2022. In 2002, the monitoring sites were mostly concentrated in California; however, there seems to be a little broader distribution throughout various regions, including both coastal and more interior areas. In 2022, the distribution seems to be more densely concentrated in certain metropolitan areas. The overlapping marks indicated that although many sites endured across two decades, others were discontinued or newly introduced. The general trend indicated that the monitoring network grew more consistent, with a larger density of stations mostly in high-population and urban areas.

Question 4 Check for any data issues such as missing or implausible values of PM 2.5 in the combined dataset. Calculate the proportion of missing/implausible values for each year and report any temporal patterns you see in these observations.

```
# Define plausibility criteria for PM2.5
# EPA daily PM2.5 values are 0-500 µg/m³
eda_all <- eda_all %>%
  mutate(
    pm25_check = case_when(
      is.na(pm2.5_avg_per_day) ~ "missing",
      pm2.5_avg_per_day < 0 ~ "implausible",
```

```

    pm2.5_avg_per_day > 500 ~ "implausible",
    TRUE ~ "valid"
  )
)

```

```

# Summarize by year
pm25_summary <- eda_all %>%
  group_by(year, pm25_check) %>%
  summarise(n = n(), .groups = "drop_last") %>%
  mutate(
    prop = n / sum(n) # proportion within each year
  ) %>%
  ungroup()

print(pm25_summary)

```

```

# A tibble: 3 × 4
  year pm25_check      n    prop
<chr> <chr>      <int>  <dbl>
1 2002  valid      15976  1
2 2022  implausible   215  0.00359
3 2022  valid      59703  0.996

```

Report:In the 2002 dataset, all 15,976 observations were genuine, with no missing or implausible values. In contrast, the 2022 dataset had a total of 59,918 observations, of which 99.6% (59,703) were valid, while 0.36% (215) were identified as implausible values. A marginal rise pattern in implausible values over time was observed. A negligible percentage (0.36%) of results were implausible, indicating modest data quality concerns over time that may warrant consideration in future epidemiological research.

Question 5 (30 points) Explore the main question of interest at three different levels of spatial resolution. Create data visualizations (e.g. boxplots, histograms, line plots, violin plots) and summary statistics that best suit each level of the analysis. Be sure to write up explanations of what you observe at each level. Level 1: State. Examine the primary question for the entire state. Level 2: County. Examine the primary question for every county in

California. Level 3: City. Restrict the data to sites in Los Angeles county and examine the primary question for every site.

```
# ensure Date is parsed and PM2.5 is clean
eda_all <- eda_all %>%
  mutate(
    Date = as.Date(Date),
    year = as.integer(year),
    pm25 = as.numeric(`pm2.5_avg_per_day`)
  ) %>%
  filter(State == "California",
    year %in% c(2002, 2022),
    !is.na(pm25),
    pm25 >= 0, pm25 <= 500) # drop implausible values
```

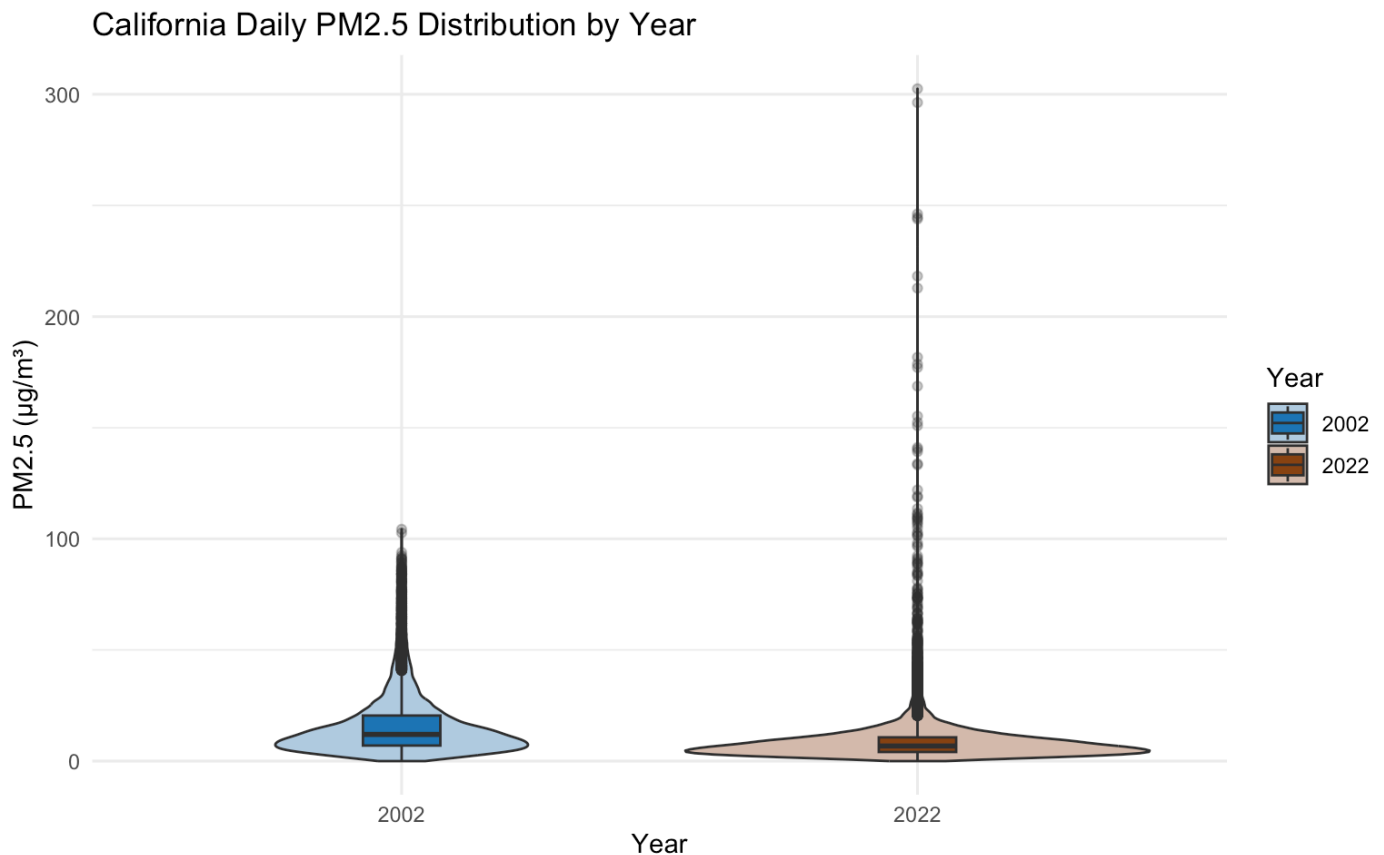
Level 1 Statewide

```
ca_state_summary <- eda_all %>%
  group_by(year) %>%
  summarise(
    mean_pm25 = mean(pm25),
    median_pm25 = median(pm25),
    p25 = quantile(pm25, 0.25),
    p75 = quantile(pm25, 0.75),
    .groups = "drop"
  )
print(ca_state_summary)
```

A tibble: 2 × 5

	year	mean_pm25	median_pm25	p25	p75
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	2002	16.1	12	7	20.5
2	2022	8.45	6.8	4.1	10.7

```
ggplot(eda_all, aes(x = factor(year), y = pm25, fill = factor(year))) +
  geom_violin(trim = TRUE, alpha = 0.4) +
  geom_boxplot(width = 0.15, outlier.alpha = 0.3) +
  scale_fill_manual(values = c("2002" = "#1f78b4", "2022" = "chocolate4")) +
  labs(title = "California Daily PM2.5 Distribution by Year",
    x = "Year", y = "PM2.5 (µg/m³)", fill = "Year") +
  theme_minimal()
```



Summary and explanation: In 2002, the average PM2.5 concentration was $16.12 \mu\text{g}/\text{m}^3$, while the median PM2.5 concentration was $12.0 \mu\text{g}/\text{m}^3$. In 2022, the average PM2.5 concentration was $8.45 \mu\text{g}/\text{m}^3$, while the median PM2.5 concentration was $6.8 \mu\text{g}/\text{m}^3$. The mean and median PM2.5 values significantly decreased from 2002 to 2022. The mean values decreased by about fifty percent (16.1 to $8.4 \mu\text{g}/\text{m}^3$) from 2002 to 2022, and the median value also declined (12.0 to $6.8 \mu\text{g}/\text{m}^3$). These patterns indicate possible decreases in population-wide exposure to fine particulate matter (PM2.5). Between 2002 and 2022, statewide PM2.5 concentrations in California had a substantial reduction, with both mean and median values decreasing by about fifty percent, and the distribution changing towards lower concentrations.

Level 2 County

```
ca_county_summary <- eda_all %>%
  group_by(County, year) %>%
  summarise(
    median_pm25 = median(pm25),
    mean_pm25   = mean(pm25),
    .groups = "drop"
```

) %>%

```
pivot_wider(names_from = year, values_from = c(median_pm25, mean_pm25))
print(ca_county_summary)
```

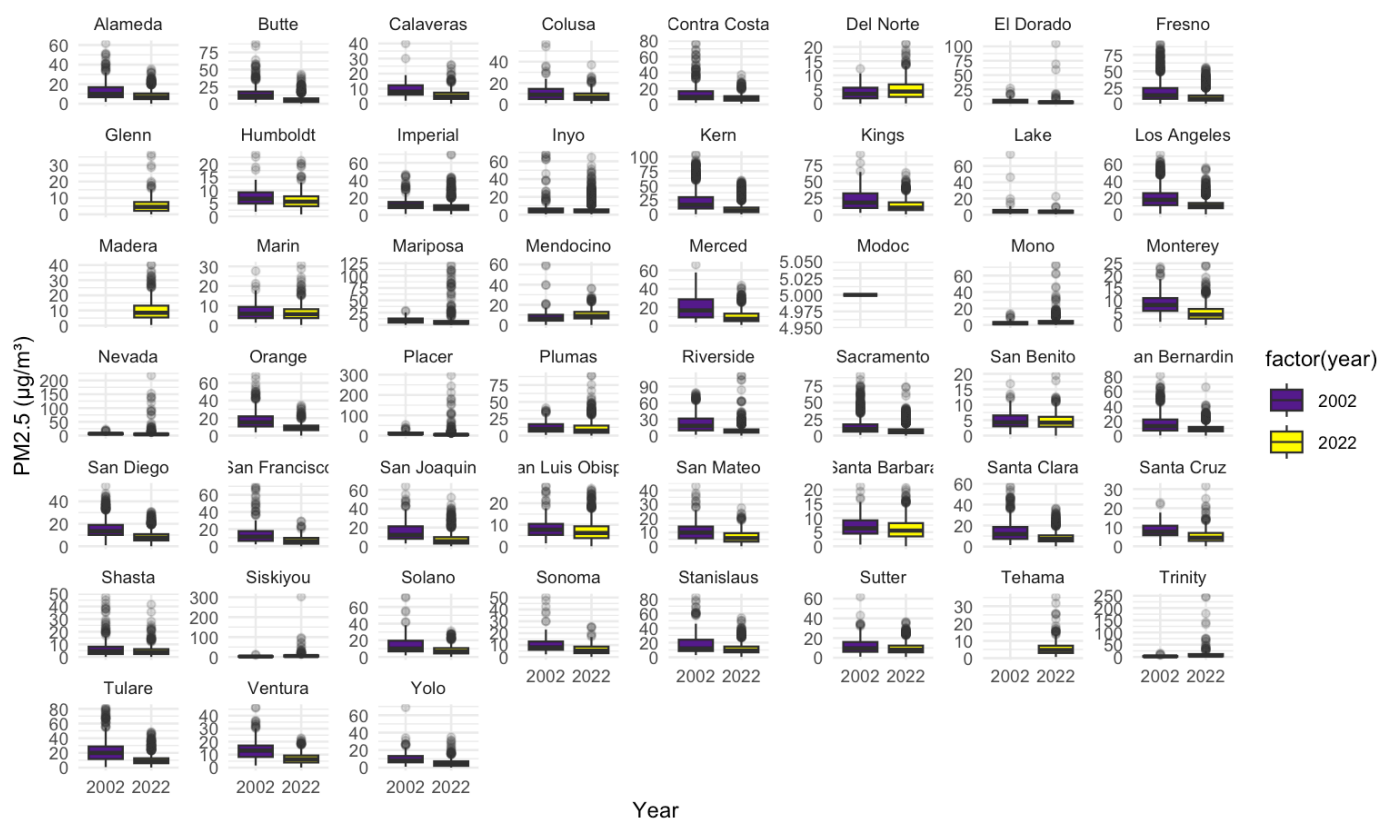
A tibble: 51 × 5

	County	median_pm25_2002	median_pm25_2022	mean_pm25_2002	mean_pm25_2022
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Alameda	10	7	14.3	8.21
2	Butte	11.5	4.5	14.8	6.26
3	Calaveras	8	5	9.9	6.04
4	Colusa	9	6.7	11.7	7.61
5	Contra Costa	9.5	7.2	15.1	8.24
6	Del Norte	3.45	4.3	3.82	5.04
7	El Dorado	3.95	2.7	4.91	4.07
8	Fresno	13.0	7.7	19.9	10.2
9	Glenn	NA	4.6	NA	5.62
10	Humboldt	6.8	5.7	7.79	6.76

i 41 more rows

```
ggplot(eda_all, aes(x = factor(year), y = pm25, fill = factor(year))) +
  geom_boxplot(outlier.alpha = 0.2) +
  facet_wrap(~ County, scales = "free_y") +
  scale_fill_manual(values = c("2002" = "purple4", "2022" = "yellow1")) +
  labs(title = "California PM2.5 by County (2002 vs 2022)",
       x = "Year", y = "PM2.5 (µg/m³)") +
  theme_minimal(base_size = 9)
```

California PM2.5 by County (2002 vs 2022)



Summary and explanation: The results indicated a general drop across counties from 2002 to 2022; counties such as Alameda, Butte, Fresno, and Los Angeles had lower median and mean PM2.5 levels in 2022 compared to 2002. The mean PM 2.5 concentration in Los Angeles decreased from 19.7 to around 11 $\mu\text{g}/\text{m}^3$, while the median declined from 17.4 to 10.3 $\mu\text{g}/\text{m}^3$. Some counties, like as Siskiyou County and Trinity County, did not see a reduction and even had an increase. In conclusion, major counties such as Los Angeles and San Francisco shown significant decreases, whereas rural and wildfire-prone regions in Northern California often demonstrated greater unpredictability in 2022. These patterns demonstrated increased interquartile ranges in 2022 for some counties, along with wildfire area.

Level 3 City

```
la <- eda_all %>%
  filter(County == "Los Angeles") %>%
  mutate(SiteName = ifelse(is.na(Local.Site.Name) | Local.Site.Name == "",
                           paste0("Site ", Site.ID),
                           Local.Site.Name))
```

```
la_site_summary <- la %>%
  group_by(SiteName, year) %>%
  summarise(median_pm25 = median(pm25), .groups = "drop") %>%
  pivot_wider(names_from = year, values_from = median_pm25)
print(la_site_summary)
```

```
# A tibble: 18 × 3
  SiteName                `2002` `2022`
  <chr>                   <dbl> <dbl>
1 Azusa                   18.7   9.65
2 Burbank                 21.6   NA
3 Compton                 NA     11.9
4 Glendora               NA     7.9
5 Lancaster-Division Street 10     7.3
6 Lebec                   4.8    3.4
7 Long Beach (North)      16.8   9.6
8 Long Beach (South)      NA     11.3
9 Long Beach-Route 710 Near Road NA     12.1
10 Los Angeles-North Main Street 19.3  10.9
11 Lynwood                19.8   NA
12 North Hollywood (NOHO) NA     13.1
13 Pasadena               17.8   7.9
14 Pico Rivera #2         NA     10.0
15 Reseda                 17.0  10.3
16 Santa Clarita          NA     8.5
```

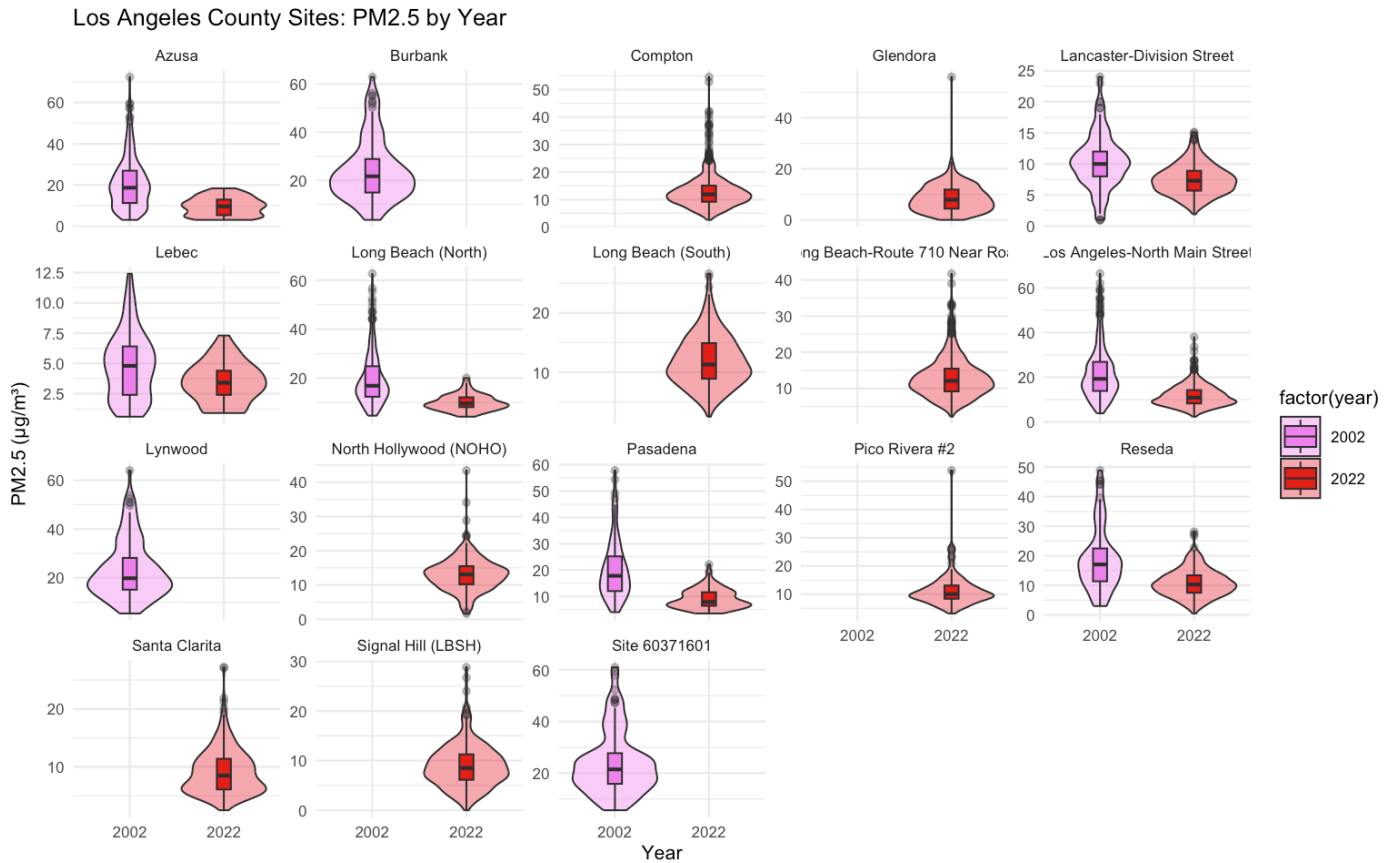
17 Signal Hill (LBSH)

NA 8.5

18 Site 60371601

21.4 NA

```
ggplot(la, aes(x = factor(year), y = pm25, fill = factor(year))) +
  geom_violin(trim = TRUE, alpha = 0.4) +
  geom_boxplot(width = 0.15, outlier.alpha = 0.3) +
  facet_wrap(~ SiteName, scales = "free_y") +
  scale_fill_manual(values = c("2002" = "violet", "2022" = "#e31a1c")) +
  labs(title = "Los Angeles County Sites: PM2.5 by Year",
       x = "Year", y = "PM2.5 ( $\mu\text{g}/\text{m}^3$ )") +
  theme_minimal(base_size = 8)
```



Summary and explanation: Numerous locations exhibited a significant decrease in PM2.5 median concentrations (e.g., Azusa: 18.7 to 9.65 $\mu\text{g}/\text{m}^3$; Long Beach North: 16.85 to 9.60 $\mu\text{g}/\text{m}^3$; Los Angeles–North Main: 19.30 to 10.90 $\mu\text{g}/\text{m}^3$). The extent of fall varies by location; for instance, the median PM 2.5 level in Reseda decreased from 17.05 to 10.3 $\mu\text{g}/\text{m}^3$, however remained comparatively high in 2022 relative to urban areas. Over fifty percent of locations had data only from a single year (e.g., Burbank and Lynwood: solely 2002; Compton, Glendora, Long Beach (South), Long Beach-Route 710 Near Road, North Hollywood, Pico Rivera #2, and Santa Clarita: exclusively 2022). This constrains comparison but still illustrates

network development, since monitoring locations have changed over time. The 2002 distributions seemed broader, with heavier tails, whereas the 2022 distributions were narrower, including fewer extreme outliers.